

Python Coding Questions

Basic Level — For Beginners & Students

Level:	Beginner / Fresher	Total Questions:	20
Topics:	Variables, Loops, Functions, Lists, Strings	Time:	2 Hours

Section 1: Variables & Data Types

Q1.

Write a Python program to declare variables of types int, float, string, and boolean. Print each variable along with its data type using the `type()` function.

Hint: Use `type(variable)` to get the data type.

Q2.

What will be the output of the following code? Explain each line.

```
x = 10
y = 3
print(x + y)
print(x - y)
print(x * y)
print(x / y)
print(x // y)
print(x % y)
print(x ** y)
```

Hint: `//` is floor division, `%` is modulus, `**` is exponent.

Q3.

Write a program that takes the user's name and age as input and prints a message like: 'Hello, Khaalid! You are 22 years old.'

Hint: Use `input()` and string formatting with f-strings.

Q4.

Swap the values of two variables `a = 5` and `b = 10` without using a third variable. Print the values before and after swapping.

Hint: Python allows: `a, b = b, a`

Section 2: Conditional Statements

Q5.

Write a program that checks if a number entered by the user is positive, negative, or zero.

Hint: Use `if`, `elif`, and `else`.

Q6.

Write a program to find the largest of three numbers entered by the user.

Hint: You can use nested `if-else` or the `max()` function.

Q7.

Write a program that checks whether a given year is a leap year or not.

Hint: A year is a leap year if divisible by 4, but not 100, unless also divisible by 400.

Q8.

What is the output of the following code? Explain the logic.

```
marks = 75
if marks >= 90:
    grade = 'A'
elif marks >= 75:
    grade = 'B'
elif marks >= 60:
    grade = 'C'
else:
    grade = 'F'
print('Grade:', grade)
```

Hint: Trace through each condition one by one.

Section 3: Loops

Q9.

Write a program to print all even numbers between 1 and 50 using a for loop.

Hint: Use range(start, stop, step) or check $i \% 2 == 0$.

Q10.

Write a program to calculate the factorial of a number entered by the user using a while loop.

Hint: factorial(5) = $5 \times 4 \times 3 \times 2 \times 1 = 120$

Q11.

Write a program to print the following pattern using nested loops: *****

Hint: Use two for loops — outer for rows, inner for stars.

Q12.

Write a program to find the sum of digits of a given number. For example, for 1234, the output should be 10.

Hint: Use modulus (%) to extract digits or convert number to string.

Q13.

Write a program to print the first 10 terms of the Fibonacci sequence.

Hint: Each term is the sum of the previous two: 0, 1, 1, 2, 3, 5, 8 ...

Section 4: Functions

Q14.

Write a function called is_prime(n) that returns True if n is a prime number, and False otherwise. Test it with a few values.

Hint: A prime number has no divisors other than 1 and itself.

Q15.

Write a function called calculator(a, b, operation) that performs addition, subtraction, multiplication, or division based on the operation string passed.

Hint: Use if-elif inside the function to select the operation.

Q16.

What will be the output of the following code? Explain the concept demonstrated.

```
def greet(name, message='Hello'):
    print(f'{message}, {name}!')

greet('Khaalid')
greet('Roshan', 'Good morning')
```

Hint: This demonstrates default parameter values in Python functions.

Section 5: Lists & Strings

Q17.

Write a program that creates a list of 5 numbers entered by the user and then prints the list sorted in ascending and descending order.

Hint: Use `list.sort()` or `sorted()`. For descending: `sorted(lst, reverse=True)`

Q18.

Write a program to count the number of vowels in a string entered by the user.

Hint: Loop through each character and check if it's in 'aeiouAEIOU'.

Q19.

Write a program to reverse a string without using the built-in `reverse()` function or slicing.

Hint: Loop through the string backwards using `range(len(s)-1, -1, -1)`.

Q20.

Given a list: `numbers = [4, 7, 2, 9, 1, 5, 8, 3, 6]` Write a program to: (a) Find the maximum and minimum values. (b) Calculate the average. (c) Remove all numbers less than 5. (d) Print the final list.

```
numbers = [4, 7, 2, 9, 1, 5, 8, 3, 6]
```

Hint: Use `max()`, `min()`, `sum()`, and list comprehension for filtering.

■ Note: Try to solve each question independently before looking up solutions. Practice is the key to mastering Python!